Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2		**g**	x	**t**	**x**	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

int max(int a, int b) {
    return (a > b) ? a : b;
}

int main() {
    char s1[100], s2[100];
    scanf("%s", s1);
    scanf("%s", s2);
    int m = strlen(s1);
    int n = strlen(s2);
    int dp[m + 1][n + 1];
    for (int i = 0; i <= m; i++)
    {
        dp[i][0] = 0;
    }
    for (int j = 0; j <= n; j++)
    {
        dp[0][j] = 0;
    }
    for (int i = 1; i <= m; i++)
    {
        for (int j = 1; j <= n; j++)
        {
            if (s1[i - 1] == s2[j - 1])
            {
                dp[i][j] = 1 + dp[i - 1][j - 1];
            }
            else
            {
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
            }
        }
```

```
36      }
37      printf("%d", dp[m][n]);
38      return 0;
39  }
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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